

In finance, time-weighted average price (TWAP) is the average price of a security over a specified time.

TWAP is also sometimes used to describe a TWAP card, that is a strategy that will attempt to execute an order and achieve the TWAP or better. A TWAP strategy underpins more sophisticated ways of buying and selling than simply executing orders en masse: for example, dumping a huge number of shares in one block is likely to affect market perceptions, with an adverse effect on the price.

Use

A TWAP strategy is often used to minimize a large order's impact on the market and result in price improvement. High-volume traders use TWAP to execute their orders over a specific time, so they trade to keep the price close to that which reflects the true market price. TWAP orders are a strategy of executing trades evenly over a specified time period. Volume-weighted average price (VWAP) balances execution with volume. Regularly, a VWAP trade will buy or sell 40% of a trade in the first half of the day and then the other 60% in the second half of the day. A TWAP trade would most likely execute an even 50/50 volume in the first and second half of the day.

Formula

TWAP is calculated using the following formula:

T

W

A

P

=

Σ

j

P

j

.

T

j

Σ

j

T

j

$$P_{\mathrm{TWAP}} = \frac{\sum_j P_j \cdot T_j}{\sum_j T_j}$$

where:

P

T

W

A

P

$$P_{\mathrm{TWAP}}$$

is Time Weighted Average Price;

P

j

$$P_j$$

is the price of security at a time of measurement

j

$$j$$

;

T

j

$\{\displaystyle T_{j}\}$

is change of time since previous price measurement

j

$\{\displaystyle j\}$

;

j

$${\displaystyle j}$$

is each individual measurement that takes place over the defined period of time.

Increased period of measurements

$$\Sigma$$

j

T

j

$$\sum_{j=1}^n T_j$$

results in a less up-to-date price.

See also

Volume-weighted average price

Slippage (finance)

== References ==